

Chinmay Burgul

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TECHNICAL SKILLS

- **Subject knowledge:** Estimation theory, Kalman Filters, NonLinear Least-Square Optimization, Computer Vision, Multi-Modal Sensor Fusion & Calibration, SLAM
- **Programming, Frameworks & Tools:** C++, Python, ROS1, ROS2, Docker

EDUCATION

- **University of Delaware** 2021 - Present
Ph.D Robotics, State Estimation & SLAM Newark, DE
- **Worcester Polytechnic Institute** 2018-2020
MSc Robotics Worcester, MA
- **SRM Institute of Science and Technology** 2012-2016
B.Tech Mechatronics Chennai, India

EXPERIENCE

- **Graduate Research Assistant** Jan 2021 - Present
University of Delaware Newark, DE
 - Developed robust state estimation and SLAM frameworks utilizing diverse multi-sensor fusion pipelines - Lidar, 3D Sonar, Visual-Inertial, Joint encoders, IMU, DVL, Multi-Body Dynamics
 - Optimized algorithmic performance for real-time execution, successfully deploying robust C++/ROS autonomy stacks directly onto physical robotic platforms
- **Graduate Research Assistant** Aug 2019 - Aug 2020
Worcester Polytechnic Institute Worcester, MA
 - Advanced research in Within-Hand Manipulation (WIHM) by formulating and developing reinforcement learning-based models for dynamic object manipulation
- **Machine Learning Intern** May 2019 - Aug 2019
Phood LLC Boston, MA
 - Designed and implemented an end-to-end machine learning framework for training object detection models and deploying them as scalable inference APIs on AWS
- **Mechatronics Engineer** Jan 2016 - Feb 2017
Entra Mechatronics Chennai, India

SELECTED PROJECTS

- **Underwater SLAM with the First Compact 3D Sonar**
Tools: [C++, ROS1, ROS2, Open3D, Aqua robot]
 - Developed a visual-3D sonar extrinsic calibration methodology and characterized sensor response on different materials
 - Developed a robust multi-sensor SLAM system integrating visual, inertial, DVL, pressure and 3D sonar measurements to deliver high-fidelity localization and mapping in underwater environments
- **Online Kinematic Determination of Legged Robots**
Tools: [C++, ROS1, ROS2, Deep Robotics Jueying Lite2 and Ghost Vision 60]
 - Developed a real-time state estimation framework for quadrupedal robots by seamlessly fusing visual, inertial, and leg kinematic measurements
 - Developed a methodology to online calibrate the leg link lengths and IMU-leg spatio-temporal offsets, validating enhanced performance on the Jueying Lite2 and Ghost Vision 60 platform
- **Efficient Multimodal Quadrupedal Autonomy: Algorithm, System and Application**
Tools: [C++, ROS1, ROS2, nvidia ORIN, Ghost Vision 60]
 - Developed a full autonomy stack for quadrupeds by integrating multimodal state estimation with an online ESDF mapping system that efficiently fuses LiDAR and RGBD data
 - Developed autonomous exploration of large-scale indoor environments by deploying the integrated planning and mapping stack directly onto the Ghost Vision 60's onboard compute

MINOR PROJECTS

- Multi-Session Mapping, Experimenting neural-nets for image detection, Online mapping of Voxels and Signed Distance Field, Deep Reinforcement Learning for Within-Hand Manipulation.

PUBLICATIONS

C=CONFERENCE, S=IN SUBMISSION

- [C.1] C. Burgul, et al. 2024. **Online Determination of Legged Kinematics**. In *IROS 2024*, pp. 9043-9049. IEEE, UAE. DOI: 10.1109/IROS58592.2024.10801595
- [C.2] C. Burgul & Y. Huang, et al. 2026. **Underwater Dense Mapping with the First Compact 3D Sonar**, *Accepted ICRA 2026*.
- [C.3] Co-authored, **Mapping Pamir: Multi-Session Visual/Inertial SLAM and 3D Reconstruction of an Underwater Shipwreck**, *Accepted ICRA 2026*.
- [S.2] C. Burgul, et al. 2026. **SVIn+: Multi Sensor SLAM with 3D Sonar, DVL, Pressure and Visual-Inertial Sensors for underwater robotics**.

OTHER SKILLS

- **Peer review:** ICRA, IROS, RAL, TRO (2022–2026)
- **Leadership Skills:** Vice-President of Robotics Graduate Student Organization(RGSO), Treasurer of Mechanical Graduate Student Organization (MEGA)